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THUNDERHACKS 2026

Official Hackathon CBN Problem Statement & Challenge Brief

★ **SILVER SPONSOR CHALLENGE** ★ | Prize Reward: **$500 CAD**

Sponsored by Canadian Bank Note.

Canadian Bank Note

Sponsor Problem Statement: Casino Style Crash Game

BACKGROUND & BUSINESS CONTEXT

Casino style crash games are among the most popular real time wagering experiences in online gaming. Players place a wager and watch a multiplier increase until the round ends at a randomly determined crash point. The challenge is balancing simplicity, excitement, fairness, and a polished user experience while maintaining a maintainable and extensible software architecture.

THE CORE PROBLEM

> *Design and build a casino style crash game using the technology stack of your choice. A working game is the foundation, but teams will be evaluated on architecture, usability, maintainability, extensibility, and overall product quality.*

Judges are looking for solutions that demonstrate thoughtful design decisions and a polished end user experience rather than a large collection of incomplete features.

MINIMUM VIABLE PRODUCT REQUIREMENTS

Your solution should include the following core capabilities:

01 Controls

Play/Place Bet and Cash Out controls.

02 Display

Display of account balance, wager amount, and current multiplier.

03 Balance Tracking

Account balance tracking.

04 Crash Point & Outcome

Randomly generated crash point and game outcome resolution.

EXPECTED GAMEPLAY FLOW

1. Player places a wager and starts a round.
2. The wager amount is deducted from the displayed balance.
3. A random crash point is generated between 1.00x and a predefined maximum multiplier.
4. The multiplier increases in real time.
5. If the player cashes out before the crash point, winnings equal wager multiplied by the current multiplier and are added to the balance.
6. If the crash occurs before cash out, the wager is lost.
7. The game resets and is available for another round.

JUDGING RUBRIC (50 POINTS TOTAL)

Category	Points	Criteria
Presentation Layer	**10**	User interface quality, responsiveness, theme consistency, feature usability, and gameplay feedback.
Backend Layer	**10**	Game logic, account management, RNG implementation, security, and reliability.
Modularity & Configurability	**10**	Code quality, maintainability, extensibility, localization, and ease of future enhancement.
Creativity & Polish	**20**	Innovation, visual appeal, player engagement, and overall impact.

OPTIONAL ENHANCEMENTS

Teams are encouraged to explore additional capabilities such as automated cash out, configurable betting options, session history, bonus rounds, analytics, accessibility features, localization support, or multiplayer concepts. These enhancements are not required but may contribute to a stronger overall submission when implemented effectively.

GLOSSARY

Auto Cash Out: A player defined multiplier threshold that automatically triggers a cash out.

RNG: Random Number Generator used to determine game outcomes.

Server Idempotence: Repeated execution of a request results in the same outcome as a single execution.

RTP: Return to Player, representing the long term average payout percentage.

★ Sponsor Award, $500 CAD Prize

The team that best addresses Canadian Bank Note's problem statement, as evaluated against the judging rubric above, will receive a **$500 CAD cash prize**. This award is judged independently of the main ThunderHacks rubric and will be presented at the closing ceremony.

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